

CSA 0378-DATA STRUCTURES

24. KRUSHKAL'S ALGORITHM

```
main.c
1 #include <stdio.h>
2
3 struct Edge {
4     int u, v, w;
5 };
6
7 int parent[10];
8
9 int find(int i) {
10     while(parent[i] != i)
11         i = parent[i];
12     return i;
13 }
14
15 void unionSet(int a, int b) {
16     parent[a] = b;
17 }
18
19 int main() {
20     struct Edge edge[5] = {
21         {0, 1, 2},
22         {0, 2, 3},
23         {1, 2, 1},
24         {1, 3, 4},
25         {2, 3, 5}
26     };
27
28     int i, count = 0, cost = 0;
29     int min, a, b, u, v;
30
31     for(i = 0; i < 4; i++)
32         parent[i] = i;
33
34     printf("Edges in MST:\n");
35
36     while(count < 3) {
37         min = 999;
38
39         for(i = 0; i < 5; i++) {
40             if(edge[i].w < min) {
41                 min = edge[i].w;
42                 a = edge[i].u;
43                 b = edge[i].v;
44             }
45         }
46
47         u = find(a);
48         v = find(b);
49
50         if(u != v) {
51             printf("%d - %d : %d\n", a, b, min);
52             cost += min;
53             count++;
54             unionSet(u, v);
55         }
56
57         for(i = 0; i < 5; i++) {
58             if(edge[i].u == a && edge[i].v == b && edge[i].w == min) {
59                 edge[i].w = 999;
60                 break;
61             }
62         }
63     }
64
65     printf("Total Cost = %d\n", cost);
66
67     return 0;
68 }
```

Output

Selected Edges:
B-C = 1
A-B = 2
B-D = 4
Total Cost = 7

=== Code Execution Successful ===